

Batch $\rightarrow \{(I_1, T_1), (I_2, T_2), \dots, (I_N, T_N)\}$

Let s_{ij} = similarity b/w Image i and caption j

$$\Rightarrow s_{ij} = \cos(I_i, T_j)$$

$$P(T_j | I_i) = \frac{\exp(s_{ij}/\tau)}{\sum_{k=1}^N \exp(s_{ik}/\tau)}$$

$$P(i|i) = \frac{e^{s_{ii}/\tau}}{\sum_{k=1}^N e^{s_{ik}/\tau}} \approx 1 \text{ (should be)}$$

↓
Probability that
Caption i is selected for Image i

In machine learning, instead of maximizing probability, we minimize loss.

$$L_i = -\log P(i|i)$$

$$= -\log \left(\frac{e^{s_{ii}/\tau}}{\sum_{k=1}^N e^{s_{ik}/\tau}} \right)$$

InfoNCE loss for image i

$$\Rightarrow L_i = - \left[\log e^{s_{ii}/\tau} - \log \left(\sum e^{s_{ik}/\tau} \right) \right]$$

$$\Rightarrow L_i = -\frac{s_{ii}}{\tau} + \log \left(\sum_{k=1}^N e^{s_{ik}/\tau} \right)$$

InfoNCE loss for image i .

↪ if $s_{ii} \gg s_{ik}$, then the summation is dominated by s_{ii}

$$\Rightarrow -\frac{s_{ii}}{\tau} + \log \left(e^{s_{ii}/\tau} + \sum_{k \neq i} e^{s_{ik}/\tau} \right) \approx 0$$

↓
minimized loss

Temperature (τ):

like a scaling factor that controls the confidence of the softmax.

$\tau \rightarrow 0$ largest similarity dominates softmax.

the large exponentials explode to ∞ and others become too small.

Here, if a wrong one is large, then the wrong captions dominate.

Here, the correct similarity must be largest, no matter by how much.

$\tau \rightarrow \infty$ Becomes constant

$\tau \rightarrow \infty \Rightarrow$ every exponent

$$\downarrow$$
$$e^{s_{ij}/\tau} = e^0 = \textcircled{1}$$

$$\Rightarrow L_i = -0 + \log N = \boxed{\log N}$$

Here, all captions get same probability.
No useful learning at all.

Gradients

$$L_{it} = -\frac{s_{ii}}{\tau} + \log \left(\sum e^{s_{ik}/\tau} \right)$$

$$\frac{\partial L_{it}}{\partial s_{ij}} = -\frac{1}{\tau} + \frac{e^{s_{ii}/\tau}}{\tau \left(\sum e^{s_{ik}/\tau} \right)}$$

$$\Rightarrow \boxed{\frac{\partial L_{it}}{\partial s_{ij}} = \frac{P(i|i) - 1}{\tau}}$$

Bad model

$$\Rightarrow P_{ii} \approx 0.2$$

$$\Rightarrow \text{gradient} = \frac{-0.8}{\tau}$$

$$\Rightarrow s_{ii} \uparrow \Rightarrow P(i|i) \uparrow$$

Good model

$$\Rightarrow P_{ii} \approx 0.99$$

$$\Rightarrow \text{gradient} = -0.01/\tau$$

$\Rightarrow$ not much change

$$\frac{\partial L_{2t}}{\partial s_{ij}} = \frac{P(j|i)}{\tau}$$

$$\therefore \frac{\partial}{\partial s_{ij}} \left(\frac{s_{ii}}{\tau} \right) \rightarrow 0$$

Suppose

$$P(j|i) \approx 0.3$$

$$\Rightarrow \frac{\partial L}{\partial s} = \frac{0.3}{\tau} > 0$$

$$\Rightarrow s_{ij} \downarrow$$

which is what
we want